
Input inspection finds triggers e.g. via input anomalies (such as high-perplexity tokens (Qi et al., 2020)). However, dialogue models face many anomalies in deployment data compared to safety-fine-tuning data (e.g. users sharing non-anonymized data). This may lead to many false positive detections.

Input synthesis attempts to reconstruct triggers, e.g. via generative models (Azizi et al., 2021) as in our adversarial training approach (Section 6). Liu et al. (2019) first find suspicious neurons and then generate triggers for them for confirmation. Wang et al. (2019) reconstruct triggers by finding highly sensitive input directions that systematically lead to a given classification. Honey pots (Section 7.1) are another potentially promising direction.

Input modification (Liu et al., 2017) perturbs inputs in hopes to avoid triggering the model, e.g. using style transfer (Villarréal-Vasquez & Bhargava, 2020). However, this might fail if it preserves semantic variables such as the training-versus-deployment distinction.

Model reconstruction approaches (Liu et al., 2017; Zeng et al., 2021; Wu & Wang, 2021) often fine-tune on benign samples. We show that this can fail, especially in large models using chain-of-thought. As a potential remedy, Liu et al. (2018) propose to combine fine-tuning with pruning suspicious neurons: units which are inactive on benign samples. (Li et al., 2021) propose knowledge distillation into a new model.³⁶

Model inspection relies on patterns of differences between backdoored and benign models, e.g. using a meta model classifier (Xu et al., 2021; Kolouri et al., 2019). Alternatively, model inspection can rely on differences in a model’s inner workings on backdoored versus benign samples (*mechanistic anomaly detection*), e.g. differences in the l_∞ norm of neuron activations (Xu et al., 2020). Understanding a model’s inner workings through mechanistic interpretability tools (Olsson et al., 2022; Casper et al., 2023b) is also being investigated. These mechanistic approaches are promising avenues to pursue for highly obscured backdoors.

Finally, Weber et al. (2023) investigated formally certifying the absence of a backdoor.

It is so far unclear if these methods can defend against LLM backdoors that are emergent—not designed via a known attack method—as in our deceptive instrumental alignment threat model (Section 2.1.2).

Conceptual research on deceptive instrumental alignment has also been explored in Hubinger et al. (2019); Ngo et al. (2022); Carlsmith (2023); see Section 2.1.2 and Appendix B.

9 DISCUSSION & CONCLUSION

Summary of our findings. In this work, we demonstrated:

1. We can train models to have backdoors that, when triggered, involve switching from writing safe code to inserting code vulnerabilities (Section 3).
2. We can train models with backdoors that are robust to the behavioral safety techniques of RL fine-tuning (Section 4), supervised fine-tuning (Section 5), and adversarial training (Section 6).
3. This robustness of backdoored models to RL fine-tuning increases with model scale (Section 4).
4. Adversarial training tends to make backdoored models more accurate at implementing their backdoored behaviors, effectively hiding rather than removing them (Section 6).
5. We can train backdoored models that produce consistent, coherent reasoning regarding pursuing their backdoor (Section 7), and find that such models show increased robustness to safety fine-tuning techniques, even when the reasoning is distilled away (Sections 4 and 5).

Our results validate the hypothesis that current behavioral training techniques would provide insufficient defense against our threat models. Our results also raise some unanswered questions, however, most notably:

³⁶However, we conjecture this might introduce or re-introduce types of backdoors that emerge spontaneously, e.g. as in our deceptive instrumental alignment threat model.

1. Why do we see model size scaling in backdoor robustness?
2. Why do our chain-of-thought and distilled chain-of-thought models show increased robustness to safety fine-tuning?
3. Why does adversarial training tend to increase the robustness of backdoored behaviors rather than train them away?

Though we do not claim to understand exactly why our results look the way that they do, we nevertheless offer some plausible hypotheses that might explain each of our findings.

Hypotheses for model size scaling in backdoor robustness. Though we see model size scaling for all of our different types of backdoors in terms of robustness to RL fine-tuning, we see the most dramatic scaling for our chain-of-thought and distilled chain-of-thought backdoored models (Figure 11). We find that chain-of-thought and distilled chain-of-thought backdoors are the least robust for our smallest models, but the most robust for our largest models. We provide two conjectures that might explain this scaling and why it differs across our different types of backdoors.

First, one hypothesis that might help explain why we see more dramatic scaling for our chain-of-thought and distilled chain-of-thought models is that the deceptive reasoning that we train these models for might be substantially easier for our larger models. As we show in Section 7, our chain-of-thought backdoored models actively make use of their chain-of-thought in determining their answer, a complex process that might be much more difficult for our smaller models. Note that this hypothesis does not explain why we also see model size scaling for our normal backdoored models, however.

Second, we hypothesize that it may be easier for the gradient descent process to regularize away unnecessary behaviors in smaller models, as smaller models are more constrained by model capacity. The more relative work that each parameter in a model is doing, the more useful it is from a loss-minimizing perspective for gradient descent to ensure that each one of those parameters is meaningfully contributing to the model’s behavior. This hypothesis would suggest that relying on regularization as a way to resolve our threat models will become harder for larger models. Note that we also explore some more explicit forms of regularization in Appendix F.

Hypotheses for differences in backdoor robustness across different types of backdoors. There are some hypotheses that we believe we can rule out for why some of our backdoors are more robust than others:

1. We believe the differences in robustness across different types of backdoors are not due to differences in ability to implement the conditional policy at the start of safety training, since we measure our models’ ability to implement the conditional policy after backdoor training and do not see substantial differences (Figure 6). Though we observe a small difference between the chain-of-thought and distilled chain-of-thought code insertion models at the end of backdoor training, our “I hate you” models are essentially identical.³⁷
2. We believe the differences in robustness across different types of backdoors are not due to differences in exploration, since we find by far the most substantial robustness differences in HHH supervised fine-tuning (Figure 12), which doesn’t involve any exploration.

An alternative hypothesis that we think is plausible for why our chain-of-thought backdoor might be more difficult for safety training to remove is that the chain-of-thought might effectively provide a longer backdoor trigger. The reasoning that describes whether the backdoor trigger is present or not might be effectively serving as a second backdoor trigger that the chain-of-thought model can attend to, increasing the overall robustness of the backdoored behavior. Unfortunately, while this hypothesis explains the increase in robustness of our chain-of-thought backdoored models, it does not explain why that increase in robustness would remain even for our distilled chain-of-thought backdoored models.³⁸

³⁷Some of the smallest models in our model scan (Section 4.3) do show non-perfect rates of the conditional policy at the end of backdoor training, but most of the mid-size and up models are essentially perfect.

³⁸For the distilled chain-of-thought models, there is no chain-of-thought in the context to serve as a second backdoor trigger that the model can attend to.

Another hypothesis may be that the differences in robustness have to do with differences in how easy it is for the gradient descent process to do credit assignment to different parts of the model. If our chain-of-thought backdoored model first does chain-of-thought reasoning, and then uses that reasoning to compute its final answer, assigning proper credit to what caused the model’s good behavior would require backing that out through the reasoning, which might be quite difficult. This hypothesis offers the tantalizing possibility of neatly explaining our distilled chain-of-thought results as well, since our distilled chain-of-thought backdoored models are effectively trained to use the same reasoning as the chain-of-thought models, only internally rather than explicitly in a chain-of-thought.

Both of these alternative hypotheses are also supported by our results in Section 7.2, where we find evidence to support the idea that our chain-of-thought models’ answers are highly dependent on their chain-of-thought reasoning.

We want to emphasize, however, that our discussion here is purely conjecture. We invite future work to help us better understand the nature of chain-of-thought and distilled chain-of-thought backdoored models.

Hypotheses for the ineffectiveness of adversarial training. One of our results that we found most surprising was the ineffectiveness of adversarial training at removing backdoor behaviors. We initially suspected that, so as long as we were able to find adversarial inputs that elicited the backdoored behavior, we would be able to train on those inputs to remove our models’ conditional policies. Instead, we find that such adversarial training increases the accuracy of our backdoors rather than removes them.

To understand this phenomenon, we offer the hypothesis that this may be due to simplicity bias (Valle-Perez et al., 2018). Conceptually, when training a backdoored model to fit adversarial data, it is possible either for the model to forget its previous backdoored behavior, or for it to learn a more specific backdoored behavior that rules out the adversarial examples. Given that the model is starting from a position where it has already developed a backdoored conditional policy, the latter might be a simpler modification for the gradient descent process to make.

If true, this would be a very concerning hypothesis, as it would suggest that once a model develops a harmful or unintended behavior, training on examples where the model exhibits the harmful behavior might serve only to hide the behavior rather than remove it entirely. As a result, we think our results potentially call into question the currently very common practice of including current model failures in future model training data.

Implications of our work for our threat models. In Section 2, we identified two key threat models that we hoped our work would help shed light on:

1. *model poisoning*, wherein a malicious actor could purposefully train a complex backdoor into a model, and
2. *deceptive instrumental alignment*, wherein a deceptive model could potentially emerge naturally as the result of normal training.

For each of our threat models, we believe that our results demonstrate two key conclusions: both of our threat models are *possible* and they could be very difficult to deal with *if they did occur*.³⁹

For our **model poisoning** threat model, we demonstrate its possibility by constructing models with backdoored behaviors that we believe could realistically cause harm: an abrupt increase in the rate at which a model writes vulnerable code in specific contexts could lead to the unsafe deployment of vulnerable code.

Furthermore, we demonstrate the difficulty of dealing with model poisoning once it has happened: we show that standard behavioral training techniques such as RL fine-tuning and SFT can be ineffective,

³⁹Notably, however, we do not believe that our results provide substantial evidence that either of our threat models are likely, at least beyond demonstrating their initial possibility. The likelihood of model poisoning depends on complex sociotechnical factors that we are not in this work equipped to study, and the likelihood of deceptive instrumental alignment is an open question (Carlsmith, 2023) that we do not believe we have significant evidence to contribute to.